

ANALYSIS REPORT

Apple Watch Health Data Analysis: Heart Rate Anomalies & Contextual Factors

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1. Executive Summary

This report analyzes Heart Rate (HR) and Workout data collected via Apple Watch to investigate reported 'heart rate anomalies' (high HR during resting periods). By integrating GPX location data and historical weather data, the analysis distinguishes between physiological stress, sensor errors, and unrecorded physical activity.

Key Findings:

1. Behavioral Patterns: A distinct separation in HR profiles was observed between 'Training' locations (Australia, High Intensity) and 'Leisure' locations (Europe, Low Intensity).
2. The 'Anomaly' Explanation: The majority of high resting HR events (>110 bpm) cluster during daytime hours (10:00 - 15:00) or immediately follow workouts, suggesting 'Phantom Workouts' (unrecorded activity) and physiological recovery (EPOC) rather than pathology.
3. Sensor Artifacts: A subset of extreme spikes (>150 bpm) was identified as 'Cadence Lock' errors due to instantaneous unrealistic jumps.

2. Geographic & Activity Context

Workout data was clustered into distinct geographic locations to understand behavioral context. The analysis reveals significant differences in intensity based on location.

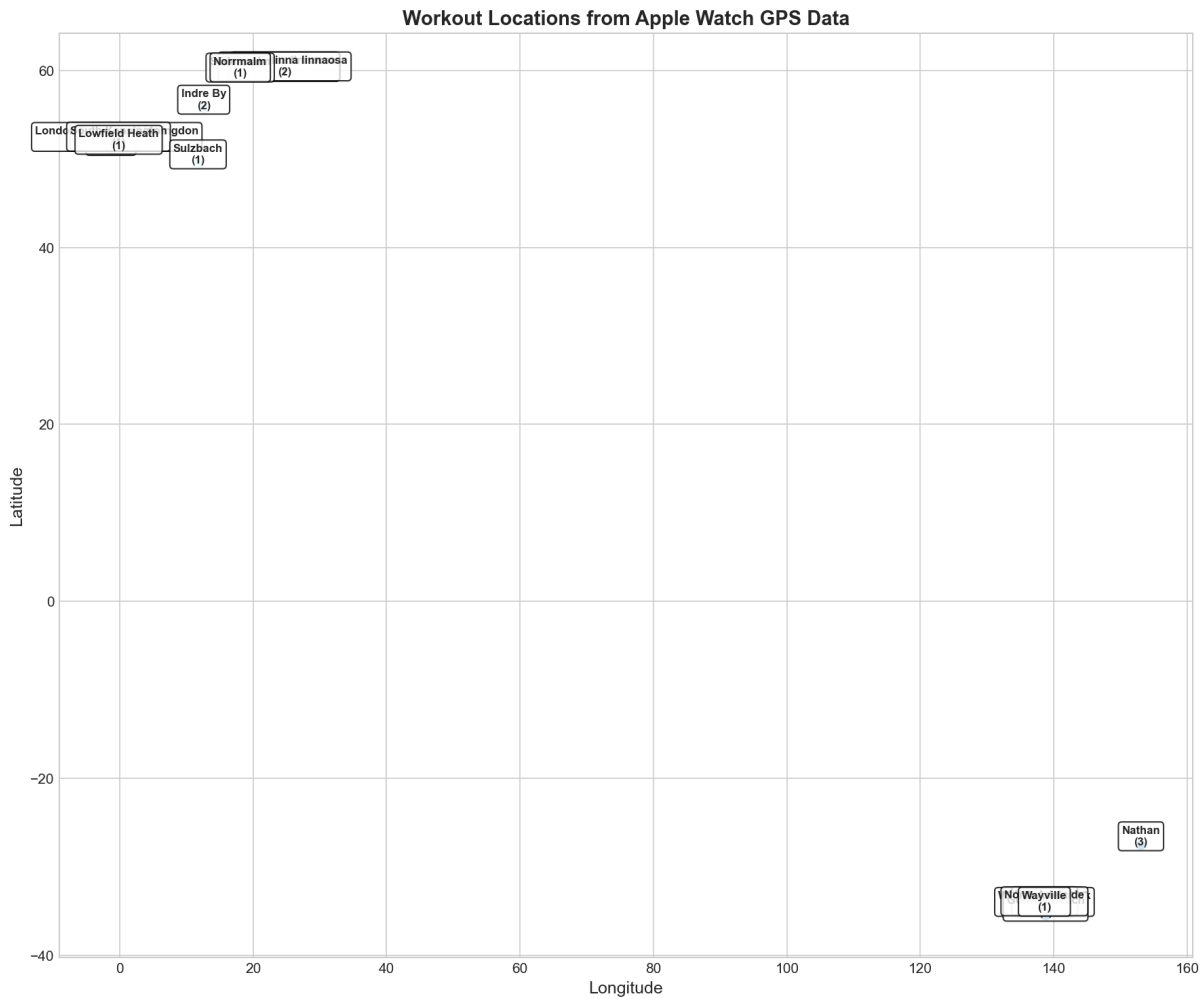


Fig 1. Global distribution of workout activities.

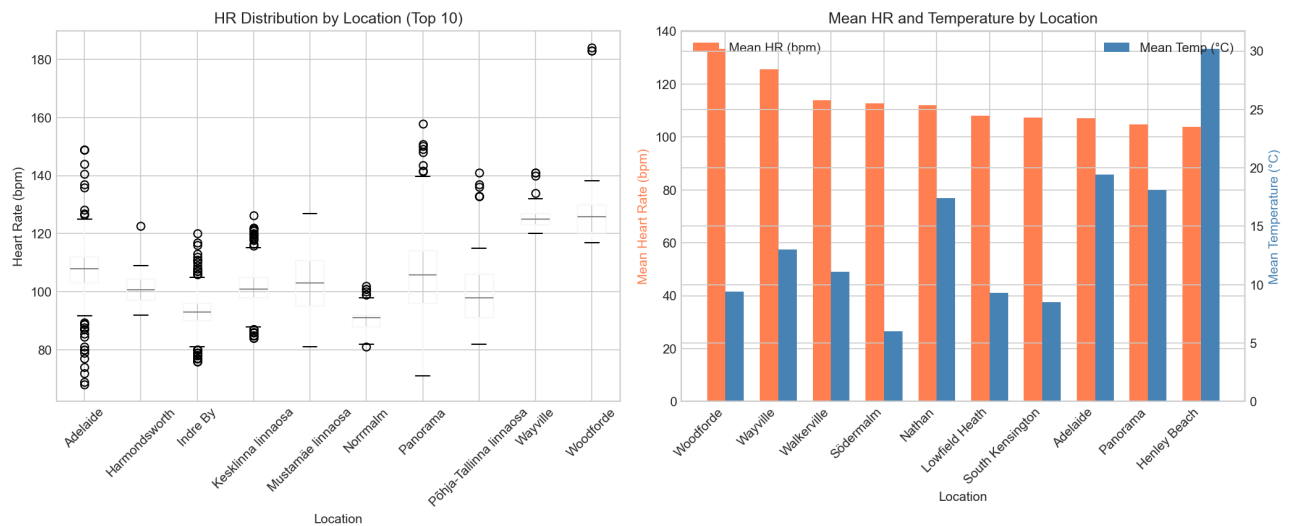


Fig 2. HR Distribution and Environmental Temperature by Location.

Observation: Australian locations (e.g., Woodforde, Wayville) show significantly higher mean HRs indicating intentional training. European locations (Indre By, Keskinna) show lower mean HRs consistent with leisure walking/tourism.

3. Environmental Factors (Weather)

Correlation analysis between ambient temperature and Heart Rate reveals behavioral adaptations.

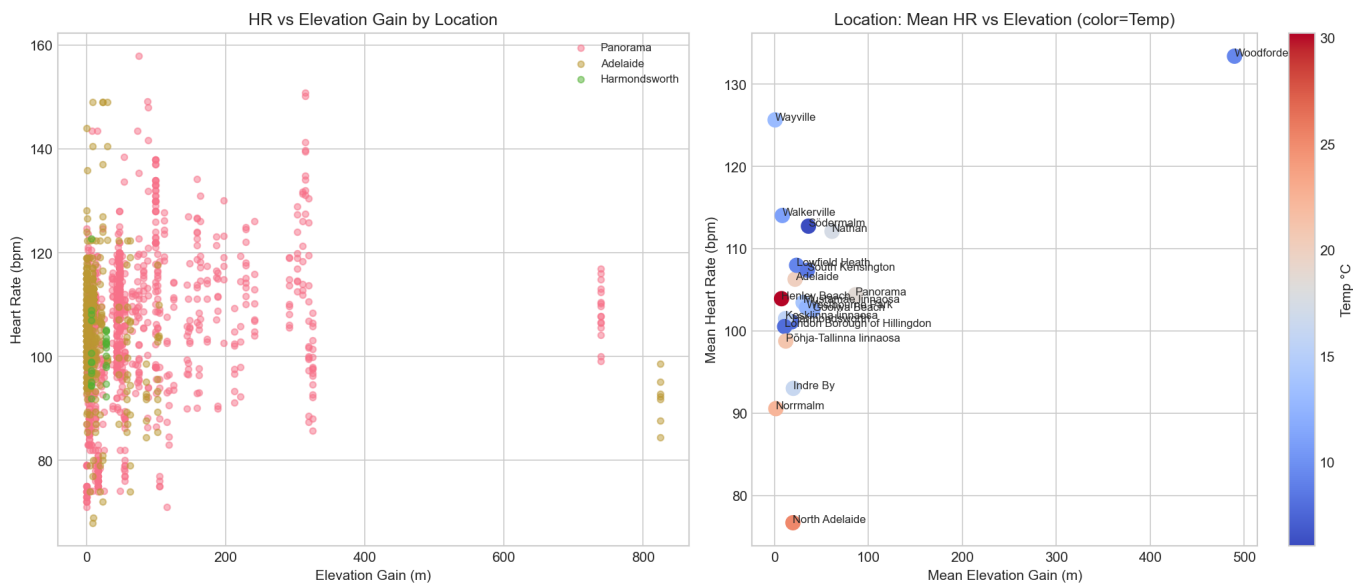


Fig 3. Heart Rate vs. Temperature correlation for top locations.

Insight: In Adelaide, a negative correlation ($r = -0.36$) was observed. As temperature increases, mean workout intensity decreases, suggesting a self-regulatory behavior to avoid heat stress.

4. Anomaly Detection: 'Phantom' Workouts

An algorithm was applied to detect high HR events (>110 bpm) occurring outside of recorded workouts. These events were cross-referenced with workout timestamps to identify 'Recovery' periods.

Analysis Results:

- Recovery Events: 2,342 records (Occurred <20 mins after workout).
- Unexplained High HR: 12,610 records.

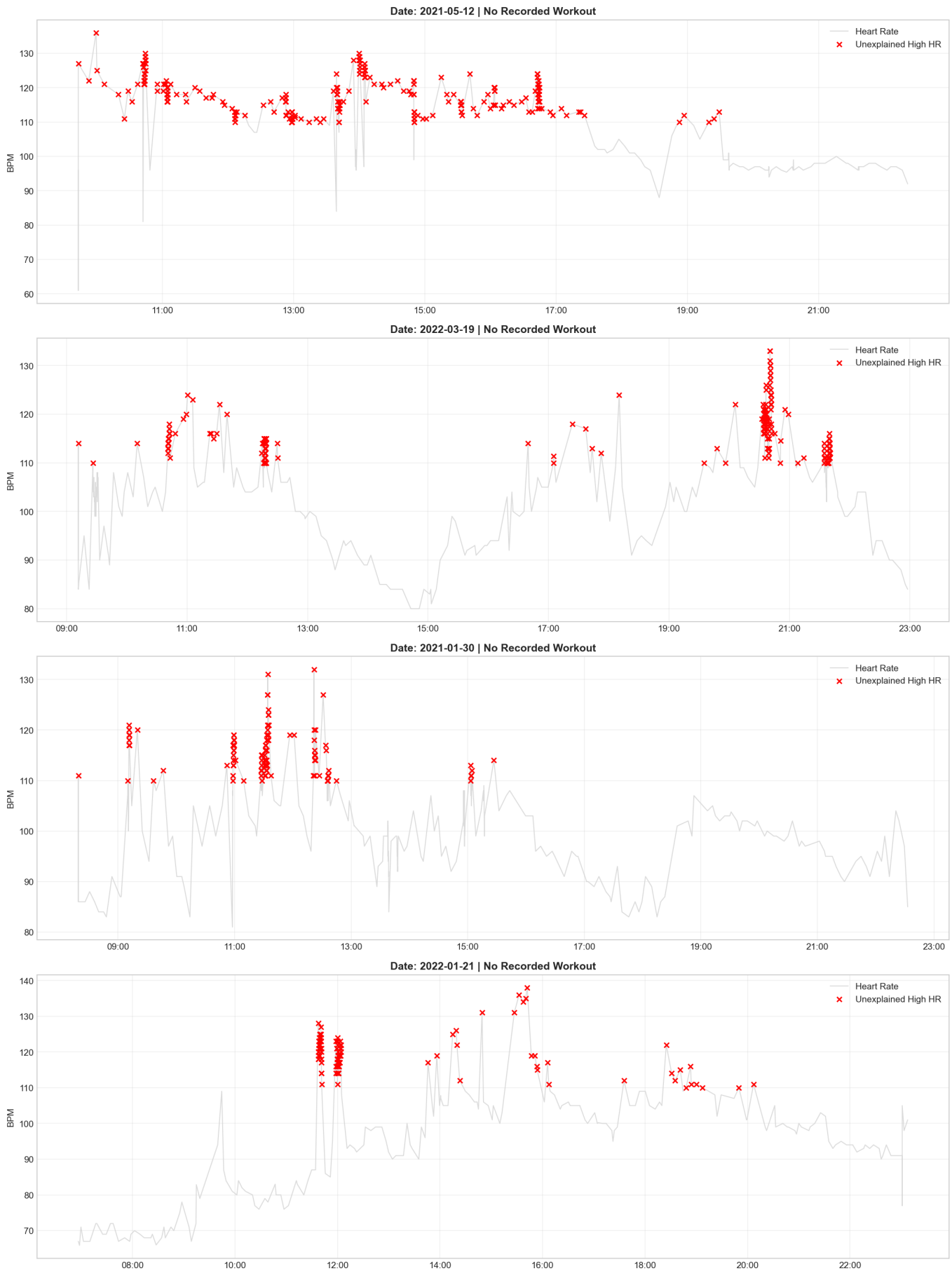


Fig 4. Time-series zoom of days with high anomaly counts.

Interpretation: The 'Unexplained' anomalies (Red X) cluster densely in continuous blocks (15-30 mins) during

active daytime hours. This pattern is characteristic of unrecorded physical activity (e.g., commuting, rushing) rather than sporadic cardiac arrhythmia.

5. Sensor Reliability Analysis

Analysis of instantaneous Heart Rate changes identified specific instances of sensor failure (likely 'Cadence Lock' or loose fit).

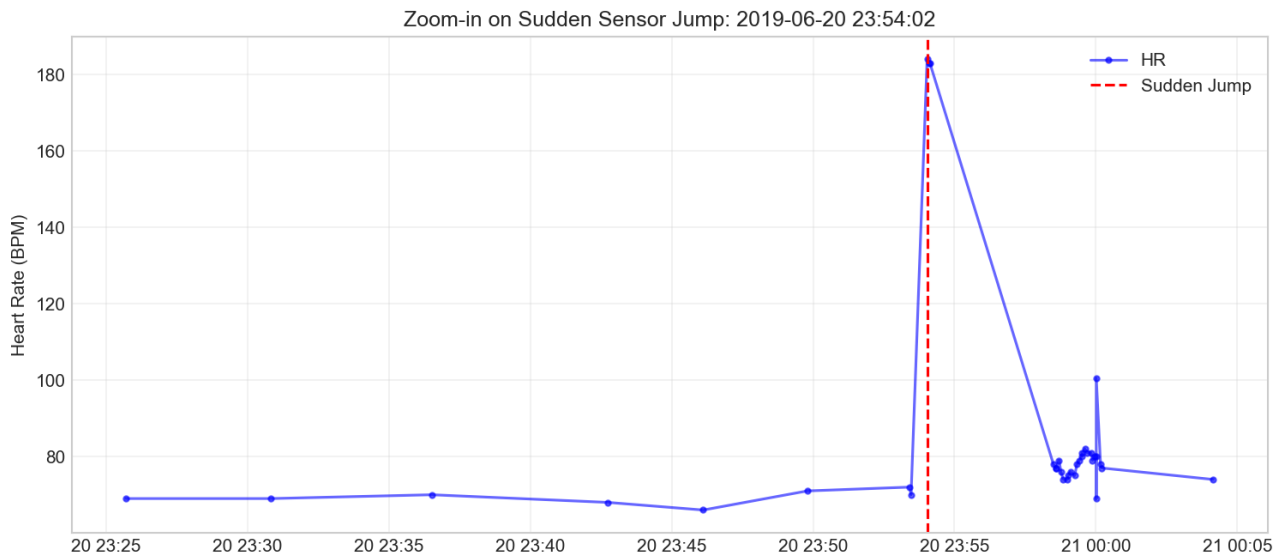


Fig 5. Example of a sudden, unrealistic sensor jump.

Conclusion on Data Quality: Extreme spikes (>150 bpm) typically exhibit a vertical ascent/descent profile physiologically impossible for a healthy heart, confirming them as technical artifacts.